

Dipan Bag

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TECHNICAL SKILLS

Languages: Python, Swift, Java, Groovy, SQL, PowerShell, JavaScript

Testing & QA: XCTest, XCUITest, Selenium, JUnit, Spock, Postman, API Testing, Regression Testing, End-to-End Testing, Test Documentation, Root-Cause Analysis, Log Analysis

CI/CD & Tools: Git, GitHub Actions, JIRA, Datadog, Docker, Studio 3T

Cloud & Systems: AWS, Azure, Google Cloud, VMware ESXi, Microsoft Hyper-V, File Systems, Distributed Systems

Databases: PostgreSQL, MongoDB, Redis, ChromaDB, Firestore, Microsoft SQL Server, NoSQL

Backend & Frameworks: SwiftUI, Spring Boot, FastAPI, Flask, Streamlit, React, LangChain, LangGraph

AI & ML: PyTorch, OpenAI, HuggingFace, RAG, LLMs, Tool-Calling, Agentic AI, Claude, GPT-5.x, Gemini

Dev Tools & IDEs: Xcode, Claude Code, Cursor, Codex, VS Code, IntelliJ IDEA, GitHub Copilot, PyCharm

EXPERIENCE

IdeaS Revenue Solutions | SDET Intern

Bloomington, MN | May 2025 – Aug 2025

- Developed end-to-end **automated tests** in **Groovy**, **JUnit**, and **Spring Boot**; automated **CI deployment** and **post-deploy validation** in **GitHub Actions**, eliminating manual verification; leveraged **GitHub Copilot** for AI-assisted development.
- Performed manual testing for **30+ JIRA-tracked** test cases in **Agile Scrum** covering negative, time-based, acceptance, performance, and **regression scenarios**; validated **API responses** and **data integrity** via **Postman**, **Studio 3T**, and **Datadog** across **MongoDB**, and **AWS**.
- Built container-agnostic **REST API** endpoints in **Spring Boot** via **Docker** to switch runtime connections between external and internal services, reducing setup for manual QA and **automated test pipelines**.

Commvault Systems | Software Development Engineer in Test

Bangalore, India | Jul 2022 – Jul 2024

- Performed automated and manual testing of features across the File Systems team across web, cloud and backend platforms using **Python**, **Selenium WebDriver** and **PowerShell**; developed expertise in **UI** and **API automation testing**.
- Planned, authored, developed and executed **test plans** and test cases; helped develop and employ **testing strategies** across features; developed **test data sets** and **scripts** to ensure adequate coverage.
- Found, documented and **tracked bugs**; worked with developers to reproduce and resolve issues; performed **regression testing**.
- Investigated automation failures through **log analysis** and **RCA** to improve repeatability and recovery workflow validation.
- Maintained and expanded the One-Touch **test automation suite by 150%** across **6 release cycles**, adding coverage for new features and updating tests; worked across **AWS**, **Azure**, **Oracle Cloud**, **VMware ESXi**, and **Hyper-V** environments.
- Debugged **70+ customer-reported issues** in **cross-functional collaboration** with developers: reproduced failures, analyzed logs, identified **root causes**, and validated fixes.

Commvault Systems | SDET Intern

Bangalore, India | Oct 2021 – Jun 2022

- Developed and maintained a **Python test automation framework**, adding coverage for new features.
- Executed test plans to validate run reliability; gained hands-on experience with **AWS**, **Windows/Unix** file systems, and **Selenium**.

SELECTED PROJECTS

DeskPilot: Local AI Desktop Assistant for macOS | *Swift, SwiftUI, EventKit, MLX, XCTest, XCUITest, Tool Calling*

- Built a local-first macOS **SwiftUI** productivity assistant combining AI chat with Notes, Dashboard summaries, Calendar events, Reminders, recent files, and configurable local model settings.
- Implemented a **tool-calling architecture** around an **MLX**-compatible local model backend, enabling deterministic tools for Notes retrieval, Calendar, Reminders, and Files over local user context.
- Developed **unit and end-to-end UI test coverage** with **XCTest** and **XCUITest**, including mocked assistant responses, coordinator-level model fakes, isolated settings tests, and a custom command-line test runner.

ProdLens: Intelligent E-Commerce Query System | *PostgreSQL, ChromaDB, LangGraph, RAG, OpenAI, LangChain*

- Built a **hybrid database** using **PostgreSQL** (Cloud SQL) and **ChromaDB** (HNSW indexing); achieved **~1ms latency** via B+ tree indexing across **1,024 products**, 122 brands, and **2,500+ reviews**.
- Reduced **API cost by 80%** via batched **LangChain ETL**; engineered **LangGraph** routing across **Text-to-SQL**, **RAG with metadata filtering**, and conversational flows; applied **HuggingFace BERT** sentiment analysis on top-rated products.

EDUCATION

University of Minnesota, Twin Cities

Minneapolis, MN | Sep 2024 – May 2026

M.S. Computer Science, GPA: **3.78/4.0**

Kalinga Institute of Industrial Technology

Bhubaneswar, India | Jul 2018 – May 2022

B.Tech. Computer Science & Engineering, GPA: **9.60/10.0**